TECHNICAL-SCIENTIFIC RESEARCH AND DEVELOPMENT (R&D) REPORT

Compliance with ABNT NBR 10719 / NBR 6022 — Edge Artificial Intelligence & Distributed Systems

NEXUS: A CASE STUDY ON THE IMPLEMENTATION OF A SOVEREIGN, AUTONOMOUS, AND PORTABLE DIGITAL COGNITIVE ORGANISM IN AN ANDROID ENVIRONMENT

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ABSTRACT

This report documents the development, architecture, and validation of NEXUS, a Digital Cognitive Organism (DCO) engineered to operate 100% locally, offline, and sovereignly on an Android mobile device (via the Termux environment). The project utilizes a compact language model (Qwen 2.5 1.5B) integrated with a Retrieval-Augmented Generation (RAG) system and an autonomous background knowledge-synthesis module. Conducted self-taught over a period of 12 months, the experimental results validate the technical feasibility of executing private generative AI on consumer mobile hardware and creating a "Cognitive Survival Kit" (FAT32 backup) ensuring structural resilience and logical immortality of the system.

Keywords: *Edge AI; RAG; Digital Sovereignty; Termux; Ollama; Human-in-the-Loop.*

1. INTRODUCTION

The contemporary evolution of Generative Artificial Intelligence reinforces a paradigm of data centralization and high computational capacity within large technology corporations. Dependence on cloud architectures (Software as a Service — SaaS) imposes severe constraints regarding data privacy, network latency, and vulnerability to continuous telemetry.

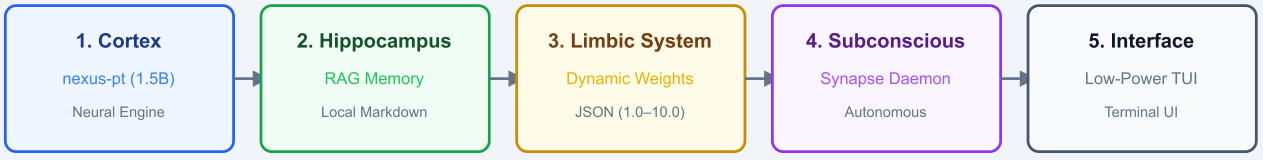
This work introduces the NEXUS system, an Edge AI architecture completely detached from external cloud servers, paid APIs, or third-party API keys. The research demonstrates the technical viability of constructing a sovereign cognitive organism running directly on an entry-level smartphone.

2. MATERIALS & METHODOLOGY

The Research and Development cycle was conducted independently between January and October 2026 by author Peterson Kennedy Silva in Betim, MG, Brazil.

- **Base Hardware:** Android Smartphone (Poco X5) with ARM64 architecture.
- **Emulation Environment:** Termux emulating a Linux subsystem for direct hardware interaction.
- **Languages & Engine:** Python 3 for orchestrating logic, Bash Scripting for system automation, and Ollama as the neural inference runtime.

FIGURE 1 — BIO-INSPIRED ARCHITECTURAL DIAGRAM OF THE NEXUS SYSTEM



3. SYSTEM ARCHITECTURE

NEXUS is structured upon five interconnected bio-inspired pillars:

3.1 Cortex (Neural Engine): Personalized `nexus-pt` model (derived from quantized Qwen 2.5 1.5B), responsible for natural language processing, instruction following, and logical reasoning.

3.2 Hippocampus (RAG Memory): Knowledge base structured in local Markdown files. The engine performs heuristic and vector retrieval prior to inference, mitigating hallucinations.

3.3 Limbic System (Weighting Mechanism): A dynamic JSON weighting table (scaling from 1.0 to 10.0) assigned to learned concepts, continually refined through operator feedback (Human-in-the-Loop).

3.4 Subconscious (Autonomous Synapse): Background process (daemon) that periodically combines random memory concepts to independently synthesize novel technical articles.

3.5 Interface (TUI): Text-based User Interface optimized to minimize RAM overhead and graphical GPU processing on mobile hardware.

4. RESULTS & VALIDATION

Experimental benchmarking validated the prototype performance across three main operational fronts:

Experimental Front	Test Scenario	Performance & Compliance Result
Offline Mode & Autonomy	Continuous execution under Airplane Mode	100% local inference at `localhost:11434` with stable memory footprint and zero external network calls.
Context Retrieval	Queries to the RAG knowledge base of technical articles	High-fidelity responses directly anchored in local documents without model hallucination.
Autonomous Synthesis	Background execution of the synapse daemon	Independent generation of coherent technical essays linking concepts like "Termux" and "Edge AI".

5. PORTABILITY & LOGICAL IMMORTALITY

System resilience and rapid deployment across unconfigured hardware are guaranteed via a streamlined recovery protocol:

- **Logical Backup:** Compression of the `~/nexus-system` directory (~80 KB), preserving interaction history, dynamic weights, and RAG documents.
- **Neural Backup:** Extraction of quantized Ollama model weights (~986 MB).
- **Autonomous Deployment Script:** Execution of `instalar_nexus.sh`, restoring the active system state in under 3 minutes using a standard 4GB FAT32 flash drive.

6. CONCLUSION

This case study proves that AI sovereignty is achievable on mobile consumer devices through architectural optimization. Peterson Kennedy Silva's experimental work provides empirical support for the "Extended Mind" thesis (Clark & Chalmers).

> *"True intelligence is not that which obeys the cloud, but that which resides in the pocket of its creator."*
>
> **— Peterson Kennedy Silva (Betim, MG, Brazil)**

BIBLIOGRAPHIC REFERENCES (ABNT NBR 6023)

CLARK, Andy; CHALMERS, David. The Extended Mind. **Analysis**, v. 58, n. 1, p. 7-19, 1998.

LEWIS, Patrick et al. Retrieval-Augmented Generation for Knowledge-Intensive NLP Tasks. **NeurIPS**, v. 33, p. 9459-9474, 2020.